

Challenge (Habanero)

- Q1.** Factor the expression $x^2 - 16$
(A) $(x + 4)(x - 4)$
(B) $x(x - 4)$
(C) $x(x + 4)$
(D) $x^2 - 4x + 16$
(E) $x^2 + 4x - 16$
- Q2.** Simplify $(a^2 - 9)(b^2 - 4)$ using difference of squares
(A) $a^2b^2 - 36$
(B) $(ab + 6)(ab - 6)$
(C) $(a + 3)(a - 3)(b + 2)(b - 2)$
(D) $a^2b^2 - 12ab + 36$
(E) $a^2 - b^2$
- Q3.** A new gaming console has $x^2 - 25$ GB of storage. Write this expression as a product of two binomials
(A) $x(x - 5)$
(B) $x(x + 5)$
(C) $(x + 5)(x - 5)$
(D) $(x - 5)(x + 25)$
(E) $x^2 + 25$
- Q4.** Which of the following is equivalent to $9x^2 - 4y^2$
(A) $(3x + 2y)(3x - 2y)$
(B) $(3x - 2y)(3x + 2y)$
(C) $9x^2 + 4y^2$
(D) $3x^2 - 4y^2$
(E) none of the above
- Q5.** Factor $64 - y^2$
(A) $8(y - 8)$
(B) $8(y + 8)$
(C) $(8 + y)(8 - y)$
(D) $(y + 8)(y - 8)$
(E) $64 + y^2$
- Q6.** A gamer scored $x^2 - 36$ points. Factor this expression
(A) $x(x - 6)$
(B) $x(x + 6)$
(C) $(x + 6)(x - 6)$
(D) $(x - 6)(x + 36)$
(E) $x^2 + 36$
- Q7.** If $x^2 - 49 = (x + a)(x - a)$, what is the value of a
(A) 7
(B) -7
(C) 49
(D) x
(E) x^2
- Q8.** A game requires $a^2 - b^2$ bytes of data to load. If $a = 8$ and $b = 3$, how many bytes of data are required
(A) 49
(B) 55
(C) $64 - 9$
(D) 65
(E) $8^2 - 3^2$

Open Ended Questions (FRQ)

- Q9.** A data plan costs

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for $x^2 - 9$ GB of data usage. If a gamer uses $x = 5$ GB, how much will they be charged. Show your work and explain your answer

- Q10.** A scoreboard has a difference of squares expression $x^2 - y^2$ where $x = 8$ and $y = 5$. Write a short explanation on what the difference of squares formula is and then apply the formula to this expression

Chef's Tips

- Ensure students understand the difference of squares formula
- Emphasize the importance of correct application of the formula
- Encourage students to double-check their work for accuracy